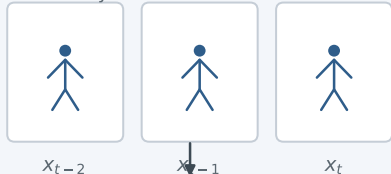


(a) Frozen observation model

pose history



Frozen GALA VAE

263-D $\rightarrow z \in \mathbb{R}^{256}$

history tokens $z_{t-k:t}$



explicit locomotion action

$a_t = \text{mean}(\text{root4})$

$(\dot{\psi}, v_x, v_z, h) \in \mathbb{R}^4$

(b) Residual dynamics p_θ

$$\hat{z}_{t+1} = z_t + \sigma(g(a_t)) \Delta$$

history (FiLM by a_t)



3-layer Transformer

self-attn on z , action as token

action gate $\sigma(g(a_t))$

$a = 0 \rightarrow \text{copy-last}$

residual unroll $H = 8$

GT actions + decoded pose loss

(c) Latent CEM planning

goal x^*

frozen encoder $\rightarrow z^*$

CEM in z

sample $a \sim \mathcal{N}(0, 0.5^2)$

rollout $\hat{z}_{1:H}$ with p_θ

cost $\|\hat{z}_H - z^*\|^2$

pop 48 · 8 iters · clip

decoder not in the loop

Hold

repeat z_t

open-loop

Decode

$\hat{x}$ for viz

/ MPJPE